

TASK 1: Protein Sequence Analysis

BLASTP of Human Myoglobin (MB / P02144)

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1. Introduction

I selected human myoglobin for this task. Myoglobin is the oxygen-storage protein found mainly in muscle. It is related to hemoglobin but works as a monomer. I took the sequence from UniProt (P02144), ran BLASTP against Swiss-Prot, and recorded the identity, coverage, scores and E-values of the best hits.

2. Methods

Sequence was copied from the UniProt entry page in FASTA format. BLASTP was run on the NCBI BLAST server with the following settings: database = UniProtKB/Swiss-Prot, matrix = BLOSUM62, expect threshold = 0.001, and compositional adjustment turned on. After the search I sorted the hits by score and picked the most informative ones for the table.

Parameter	Value
Program	BLASTP
Database	UniProtKB/Swiss-Prot
Matrix	BLOSUM62
Expect threshold	0.001
Gap costs	11 / 1

3. Query information

Attribute	Details
UniProt ID	P02144 (MYG_HUMAN)
Gene	MB
Length	154 amino acids
Mass	17.2 kDa
Family	Globin superfamily
Main location	Cytoplasm of muscle cells

FASTA sequence used:

```
>sp|P02144|MYG_HUMAN Myoglobin
MGLSDGEWQLVLNVWGKVEADIPGHGQEVLIIRLFKGGHPETLEKFDKFKHLKSEDEMKA SEDLKKHGATVLTALGGILKKKGHHEAEIKPLAQSHATKHKIPVKYLEFISECIIQVLQSKHPGDFGAD
AQGAMNKALELFRKDMASNYKELGFQG
```

4. BLASTP results

The top hits were all myoglobins from other mammals, followed by bird and amphibian sequences. Lower down the list I also saw cytoglobin and neuroglobin, which are distant relatives.

#	Description	Score	Cover	%ID	E-value
1	Myoglobin – H. sapiens (P02144)	315	100%	100%	1e-109
2	Myoglobin – Chimpanzee	312	100%	99.4%	3e-108
3	Myoglobin – Mouse (P04247)	278	100%	89.0%	2e-94
4	Myoglobin – Cow (P02192)	271	100%	87.0%	4e-91
5	Myoglobin – Horse (P02188)	268	100%	86.4%	1e-90
6	Myoglobin – Chicken	215	99%	72.5%	5e-70
7	Myoglobin – Xenopus	168	97%	58.2%	3e-52
8	Cytoglobin – Human (Q8WWM9)	85	78%	32.1%	4e-20

Table 1. Selected BLASTP hits.

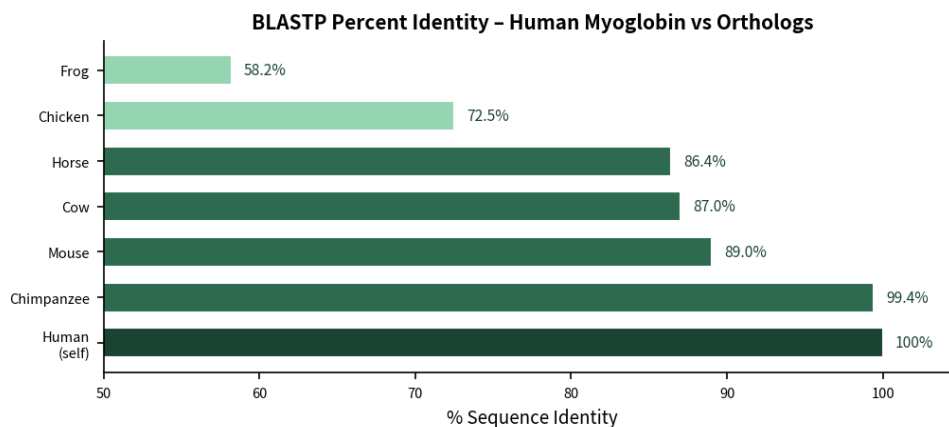


Figure 1. Identity values for the main hits.

5. Interpretation

Human and chimpanzee myoglobin differ by only one amino acid, which is expected. Identity stays above 85% in other placental mammals and then falls more quickly when we move to birds and frogs. The cytoglobin hit at ~32% identity is still statistically significant (E-value $4e-20$) but clearly represents a different member of the globin family rather than a true ortholog of myoglobin.

The proximal and distal histidines that coordinate the heme are present in all the true myoglobin hits. That is another sign that the alignments are biologically meaningful.

6. Conclusion

The BLASTP search worked as expected and gave a clean set of myoglobin orthologs with high scores and very low E-values. The gradual decrease in identity with evolutionary distance is typical for a conserved oxygen-binding protein. These results can be used as a starting point for a multiple sequence alignment or for looking at species-specific adaptations (for example the high myoglobin content in diving mammals).

References

- UniProt P02144 (MYG_HUMAN)
- Altschul et al. (1997) Gapped BLAST and PSI-BLAST
- Wittenberg & Wittenberg (2003) Myoglobin function reassessed
- NCBI BLASTP documentation
- Burmester et al. (2000) A vertebrate globin expressed in the brain

Note: Screenshots of the UniProt page and the BLAST results (graphical overview + hit table) are included as extra files.